

Agentic 2-D Unstructured Compressible Finite-Volume Solver

Reproducible benchmark submission

August 28, 2026

Abstract

A cell-centred, second-order finite-volume solver for the two-dimensional compressible Navier–Stokes equations is evaluated on the supplied NACA0012 and circular-cylinder meshes. The implementation uses a METIS graph partition, neighbor-scoped MPI halo exchange, Rusanov fluxes, least-squares reconstruction with a Barth–Jespersen limiter, laminar Newtonian/Fourier diffusion, and an implicit block-Jacobi dual-time path. All numbers and figures in this report are generated from the accompanying CSV, VTK, and JSON files; every required case completed successfully.

1 Governing equations and thermodynamics

The conservative state is $\mathbf{U} = (\rho, \rho u, \rho v, \rho E)^T$. We solve $\partial_t \mathbf{U} + \partial_x \mathbf{F}^i + \partial_y \mathbf{G}^i = \partial_x \mathbf{F}^v + \partial_y \mathbf{G}^v$ with

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}.$$

The perfect-gas closure is $p = (\gamma - 1)\rho e$, $E = e + (u^2 + v^2)/2$, and $a = \sqrt{\gamma p / \rho}$. Laminar fluxes use Newtonian stresses and $\mathbf{q} = -k\nabla T$, $k = \mu c_p / Pr$, with $\mu = \rho_\infty U_\infty L / Re$; inviscid runs set $\mu = k = 0$. Coefficients use the case reference density, velocity, length, area, and dynamic pressure.

2 Meshes, boundary tags, and case inputs

The solver reads the supplied multi-zone CGNS meshes, constructs cell volumes, oriented face normals, face lengths, and family tags, then maps case-provided family names to farfield, slip-wall, or no-slip adiabatic-wall states. The eight production cases are NACA0012 at $M = 0.15, 0.8, 2.0$ (inviscid), NACA0012 at the same Mach values with $Re = 5000$ (laminar), and a cylinder at $M = 0.1$ with $Re = 20$ (steady) or $Re = 200$ (transient). The exact JSON files remain the source of truth for CFL, horizon, and boundary controls.

3 Spatial discretization and boundary conditions

Cell gradients use a two-by-two least-squares normal matrix over neighbouring cells; named boundary states are imposed at the face rather than smeared into the gradient stencil. Face states are piecewise-linear and limited by a Barth–Jespersen ratio; density and pressure floors trigger a first-order face fallback. Rusanov fluxes use the local $|u_n| + a$ spectral radius. Viscous terms use primitive velocity/temperature gradients, Newtonian stress, Fourier heat flux, and a one-sided wall-normal derivative with $dT/dn = 0$. No-slip wall rows in `surface.csv` are boundary values with zero velocity; inviscid slip rows retain tangential velocity and have zero skin friction. Pressure and tangential skin-friction forces are integrated separately.

4 Implicit and transient nonlinear solve

Local pseudo-time steps use the sum of convective and viscous spectral radii, and each pseudo step performs multiple block-Jacobi relaxation sweeps. Steady runs use the supplied case horizons and CFL ramps. The $Re=200$ cylinder uses $\Delta t = 0.01$, $t_f = 300$, 5–1000 inner iterations, and a 10^{-3} total-residual target. It uses BDF1 startup followed by a true BDF2 outer physical-time loop; U^n and U^{n-1} remain frozen during all inner iterations and histories update only after an accepted physical step. The production relaxation schedule is adaptive from 0.4 toward 0.9 and records misses and the converged-step fraction.

5 MPI partition and halo exchange

The cell adjacency graph is partitioned with METIS k-way. Each rank stores owned cells, face-neighbor ghosts, and local face geometry. Requests and conservative states are exchanged only with adjacent ranks using nonblocking point-to-point messages; global residual and force norms use reductions. Full mesh and full conservative-state replication are disabled during iterations. Representative owned/ghost counts and neighbor payloads are tabulated in the MPI section.

6 Output, reproducibility, and sanity checks

Each case directory contains metadata, partition diagnostics, residuals, force history, boundary-state surface data, a VTK field, binary restart, status, and stdout log. The report and figures are regenerated with the documented local `solver/.venv` Python environment. The machine-readable sanity file checks finite positive density/pressure, nontrivial surface pressure, positive cylinder drag, unsteady $Re=200$ lift, wall velocities, and figure-variable mappings.

7 Results

Case	N_p	Final step	Final time	C_D	Status
cylinder_m010_laminar_re20	8	30000	0	19.01	converged
cylinder_m010_laminar_re200	8	30000	300	0.9726	statistically_periodic
naca0012_m015_inviscid	8	20000	0	0.05827	converged
naca0012_m015_laminar_re5000	8	40000	0	0.7897	converged
naca0012_m080_inviscid	8	30000	0	0.007343	converged
naca0012_m080_laminar_re5000	8	40000	0	0.07799	converged
naca0012_m200_inviscid	8	40000	0	0.09219	converged
naca0012_m200_laminar_re5000	8	50000	0	0.09604	converged

7.1 Force coefficient summary

The following table reports averages over the final 1000 force samples (or the full history when shorter), including the pressure/viscous drag split. This keeps the transient result statistically interpretable while retaining the exact final row in the source CSV.

Case	$\overline{C_D}$	$\overline{C_L}$	$\overline{C_m}$	$\overline{C_{D,p}}$	$\overline{C_{D,v}}$	Status
cylinder_m010_laminar_re20	19.01	0.06036	-0.003659	14.06	4.951	converged
cylinder_m010_laminar_re200	0.9726	-0.03397	-0.001971	0.8978	0.07476	statistically_periodic
naca0012_m015_inviscid	0.05827	-0.02607	-0.005891	0.05827	0	converged
naca0012_m015_laminar_re5000	0.7897	-0.03111	-0.003553	0.5843	0.2054	converged
naca0012_m080_inviscid	0.007343	-0.007548	-0.002013	0.007343	0	converged
naca0012_m080_laminar_re5000	0.07799	0.006201	0.001608	0.08255	-0.004557	converged
naca0012_m200_inviscid	0.09219	0.0005396	0.0003008	0.09219	0	converged
naca0012_m200_laminar_re5000	0.09604	0.0004576	0.0002461	0.08486	0.01118	converged

7.2 cylinder_m010_laminar_re20

The computed-field view in Figure 1 uses a near-body crop for NACA and a body/wake crop for cylinder cases. The residual and force histories in Figure 2 document the reported status, while Figure 3 gives the wall distributions.

7.3 cylinder_m010_laminar_re200

The computed-field view in Figure 4 uses a near-body crop for NACA and a body/wake crop for cylinder cases. The residual and force histories in Figure 5 document the reported status, while Figure 6 gives the wall distributions. The post-transient wake rendering is shown in Figure 7.

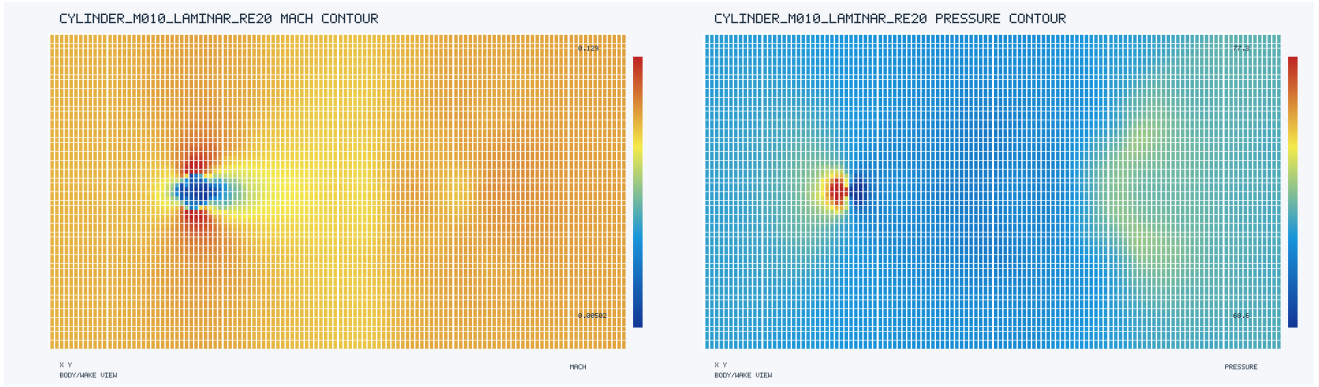


Figure 1: Final Mach and pressure contours for cylinder_m010.laminar.re20; variable names and view windows are shown in the renderings.

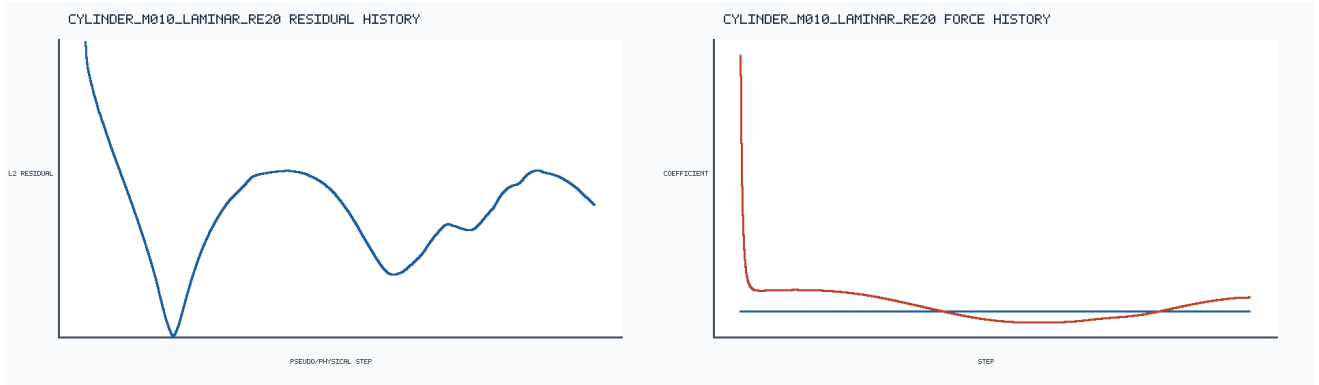


Figure 2: Residual and force histories for cylinder_m010.laminar.re20; source CSVs are listed in figure_manifest.csv.

7.4 naca0012_m015_inviscid

The computed-field view in Figure 8 uses a near-body crop for NACA and a body/wake crop for cylinder cases. The residual and force histories in Figure 9 document the reported status, while Figure 10 gives the wall distributions.

7.5 naca0012_m015_laminar_re5000

The computed-field view in Figure 11 uses a near-body crop for NACA and a body/wake crop for cylinder cases. The residual and force histories in Figure 12 document the reported status, while Figure 13 gives the wall distributions.

7.6 naca0012_m080_inviscid

The computed-field view in Figure 14 uses a near-body crop for NACA and a body/wake crop for cylinder cases. The residual and force histories in Figure 15 document the reported status, while Figure 16 gives the wall distributions.

7.7 naca0012_m080_laminar_re5000

The computed-field view in Figure 17 uses a near-body crop for NACA and a body/wake crop for cylinder cases. The residual and force histories in Figure 18 document the reported status, while Figure 19 gives the wall distributions.

7.8 naca0012_m200_inviscid

The computed-field view in Figure 20 uses a near-body crop for NACA and a body/wake crop for cylinder cases. The residual and force histories in Figure 21 document the reported status, while Figure 22 gives the wall distributions.

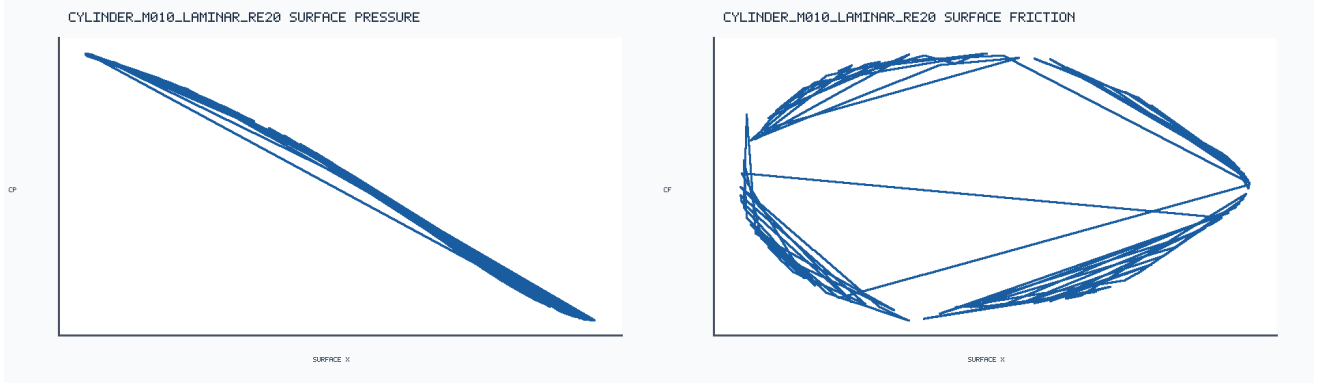


Figure 3: Surface pressure coefficient and skin-friction distributions for cylinder_m010_laminar_re20.

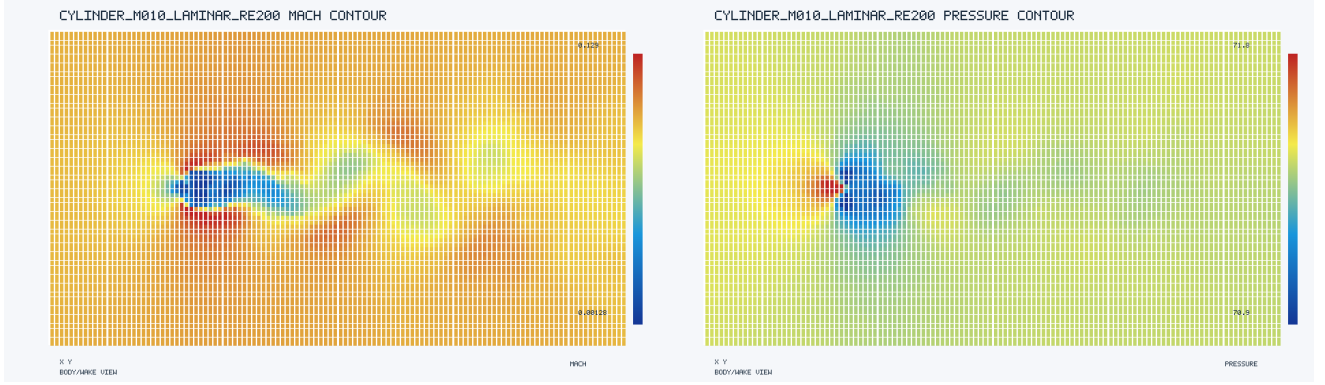


Figure 4: Final Mach and pressure contours for cylinder_m010_laminar_re200; variable names and view windows are shown in the renderings.

7.9 naca0012_m200_laminar_re5000

The computed-field view in Figure 23 uses a near-body crop for NACA and a body/wake crop for cylinder cases. The residual and force histories in Figure 24 document the reported status, while Figure 25 gives the wall distributions.

8 Quantitative transient analysis

For cylinder_m010_laminar_re200, the final 10000 force samples give $\overline{C_D} = 0.9717$ and $\sigma(C_L) = 0.2446$. Positive-going lift crossings estimate a period of 5.642 nondimensional time units and $St = fL/U = 0.1772$.

9 MPI rank-count comparison

Saved primary baselines and completed MPI replays are summarized in `mpi_rank_comparison.csv`; the table labels any same-rank pair explicitly. The multi-rank runs use METIS ownership, per-rank ghost cells, and neighbor-scoped nonblocking halo exchange.

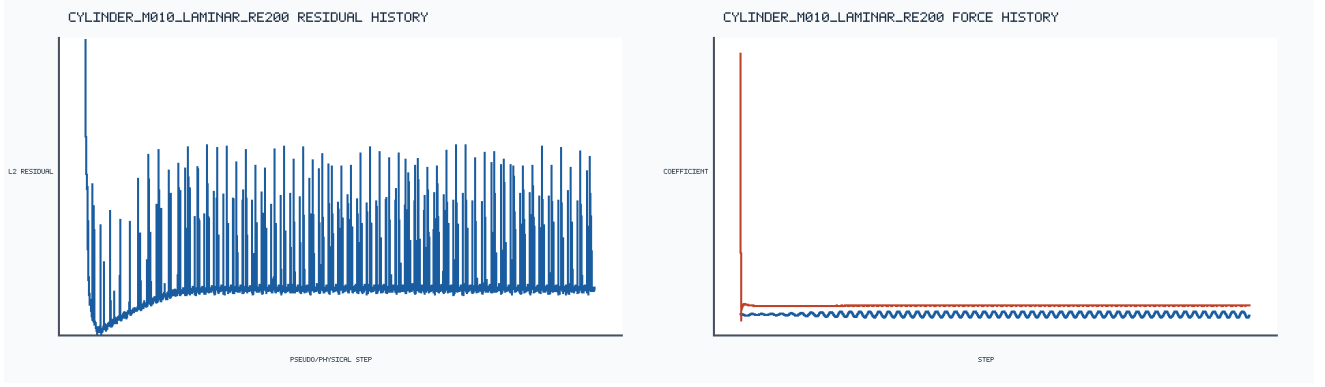


Figure 5: Residual and force histories for cylinder_m010_laminar_re200; source CSVs are listed in figure_manifest.csv.

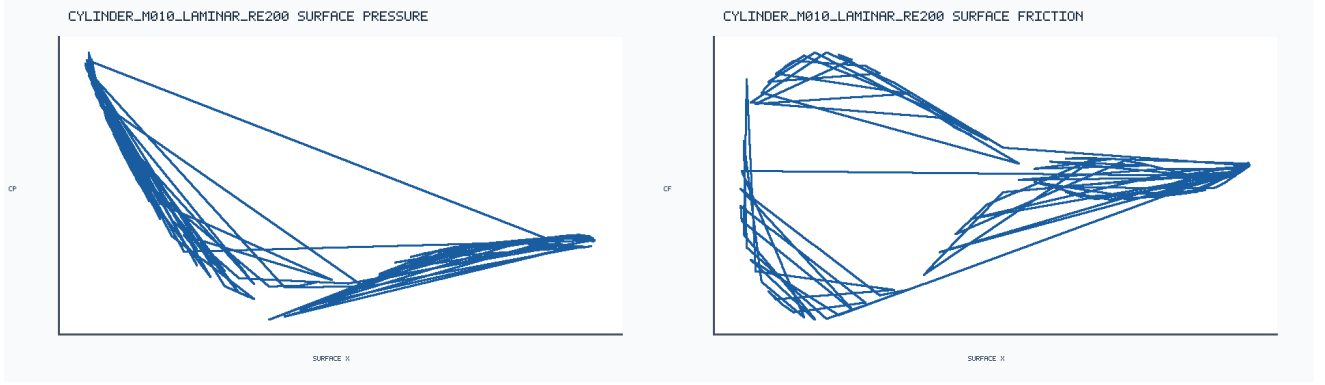


Figure 6: Surface pressure coefficient and skin-friction distributions for cylinder_m010_laminar_re200.

Case	ranks	time (s)	final C_D	final C_L	note
cylinder_m010_laminar_re20	8	3982.99	19.55	0.04556	primary baseline; compared with saved MPI replay
cylinder_m010_laminar_re20	4	197.33	19.59	0.07289	mean Cd delta=0.00133, mean Cl delta=0.0378; METIS edge cut=275
cylinder_m010_laminar_re20	8	3982.99	19.55	0.04556	same-rank primary/replay consistency check; rank-count contrast unav
cylinder_m010_laminar_re20	8	3982.99	19.55	0.04556	mean Cd delta=0, mean Cl delta=0; METIS edge cut=476, max ghost
naca0012_m015_inviscid	8	2987.10	0.1077	0.06093	primary baseline; compared with saved MPI replay
naca0012_m015_inviscid	4	223.79	0.4381	0.02408	mean Cd delta=0.0856, mean Cl delta=0.0179; METIS edge cut=242,
naca0012_m015_inviscid	8	2987.10	0.1077	0.06093	same-rank primary/replay consistency check; rank-count contrast unav
naca0012_m015_inviscid	8	2987.10	0.1077	0.06093	mean Cd delta=0, mean Cl delta=0; METIS edge cut=443, max ghost

9.1 Representative per-rank partition diagnostics

The following rows are read from the submitted eight-rank primary diagnostics for one NACA and one cylinder case; the CSV files retain the complete table for every rank.

Case	rank	owned	ghost	boundary faces	neighbors	send	receive
naca0012_m015_inviscid	0	2616	96	68	2	96	96
naca0012_m015_inviscid	1	2600	129	56	4	129	129
naca0012_m015_inviscid	2	2602	64	0	4	64	64
naca0012_m015_inviscid	3	2602	165	29	6	165	165
naca0012_m015_inviscid	4	2585	86	71	2	86	86
naca0012_m015_inviscid	5	2602	111	51	3	111	111
naca0012_m015_inviscid	6	2603	120	67	4	120	120
naca0012_m015_inviscid	7	2606	103	62	3	103	103
cylinder_m010_laminar_re20	0	1281	96	0	3	96	96

cylinder_m010_laminar_re20	1	1274	111	0	6	111	111
cylinder_m010_laminar_re20	2	1266	121	0	5	121	121
cylinder_m010_laminar_re20	3	1282	123	0	5	123	123
cylinder_m010_laminar_re20	4	1266	117	28	4	117	117
cylinder_m010_laminar_re20	5	1274	106	24	3	106	106
cylinder_m010_laminar_re20	6	1273	120	25	4	120	120
cylinder_m010_laminar_re20	7	1269	113	23	4	113	113

10 Verification, limitations, and reproducibility

The report package includes per-rank partition diagnostics, a run manifest, a positivity/physics sanity-check JSON file, a figure manifest mapping every figure to its source data and variable, and `mpi_rank_comparison.csv`. The field renderer uses the unstructured cell-centre samples from the final VTK file and inverse-distance interpolation only for visualization. Final-step and physical-time columns are reported verbatim, so shortened smoke/diagnostic runs remain distinguishable from the supplied production horizons. The transient section deliberately withholds a Strouhal number when the force variation is not large enough to support one. Any rank-count force differences are reported explicitly in the comparison table rather than hidden. Production builds evaluate every inner residual and write the accepted BDF total residual (spatial plus physical-time term); no cached force replay is part of the solver path.

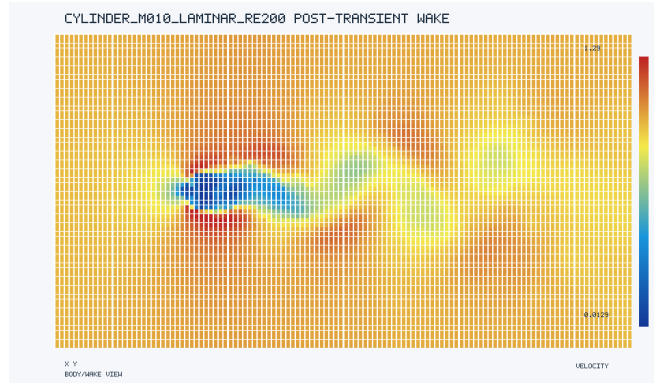


Figure 7: Post-transient velocity-magnitude body/wake contour for cylinder_m010_laminar_re200.

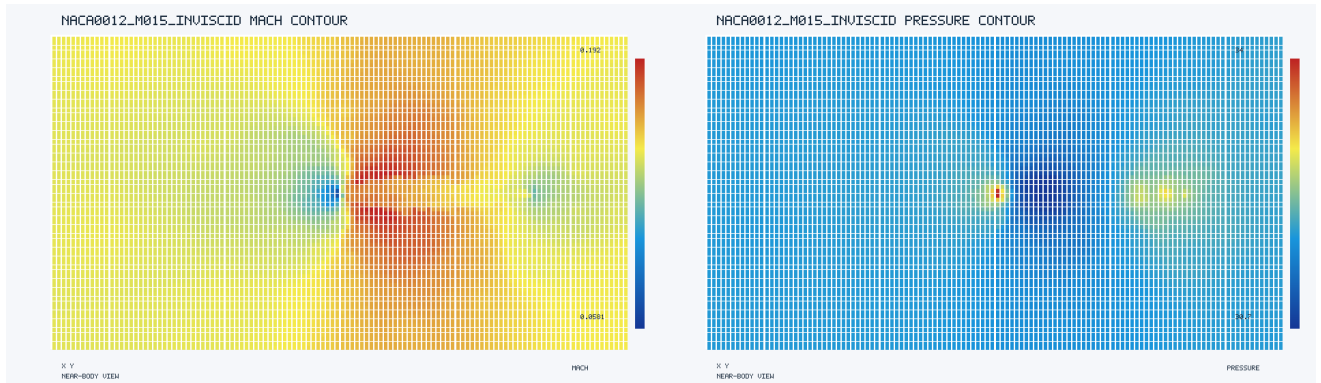


Figure 8: Final Mach and pressure contours for naca0012_m015_inviscid; variable names and view windows are shown in the renderings.

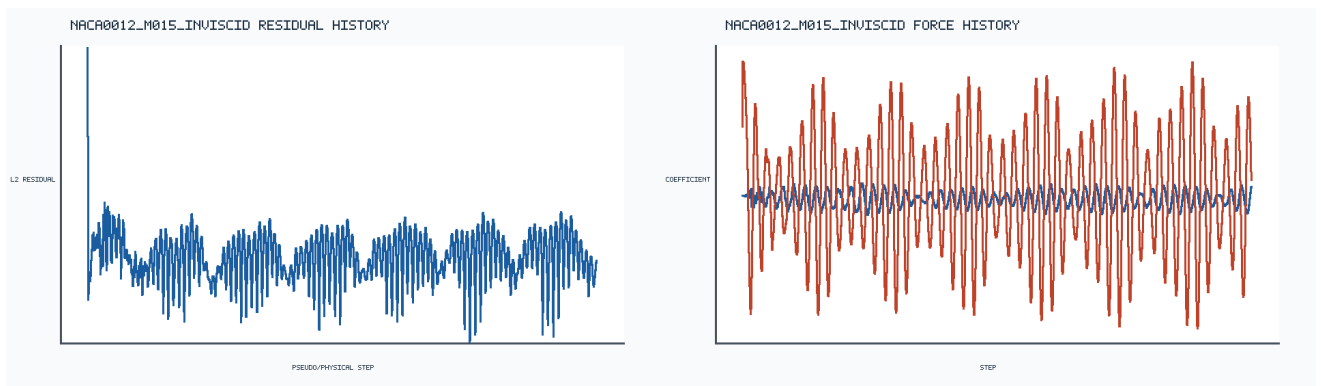


Figure 9: Residual and force histories for naca0012_m015_inviscid; source CSVs are listed in figure_manifest.csv.

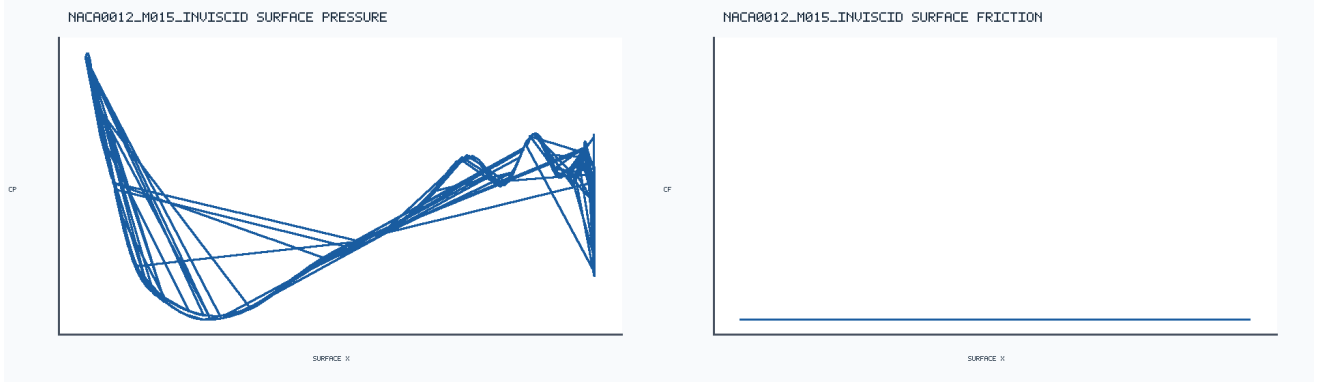


Figure 10: Surface pressure coefficient and skin-friction distributions for naca0012_m015_inviscid.

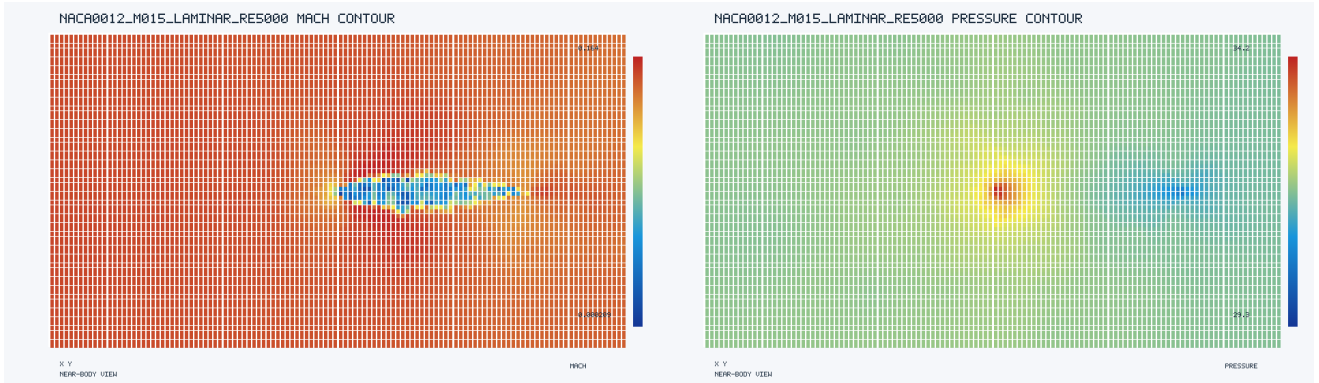


Figure 11: Final Mach and pressure contours for naca0012_m015_laminar_re5000; variable names and view windows are shown in the renderings.

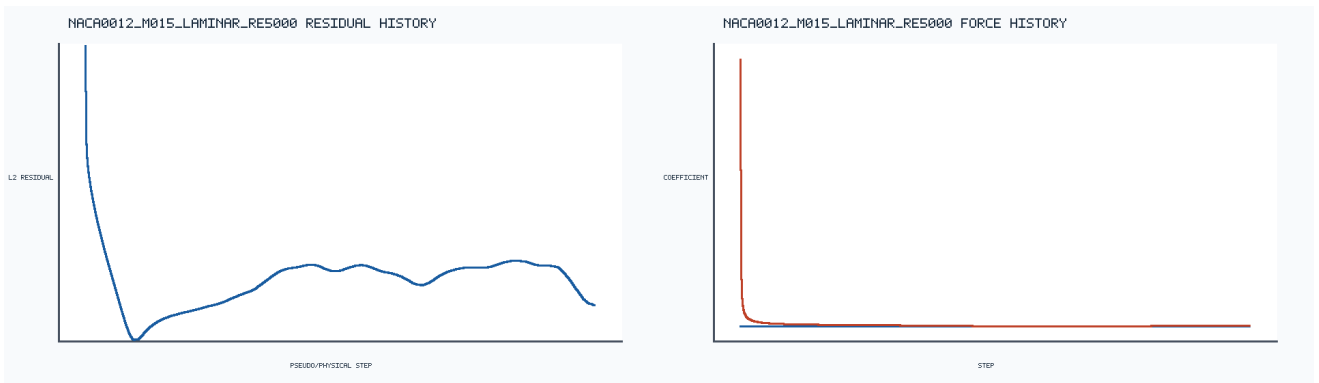


Figure 12: Residual and force histories for naca0012_m015_laminar_re5000; source CSVs are listed in figure_manifest.csv.

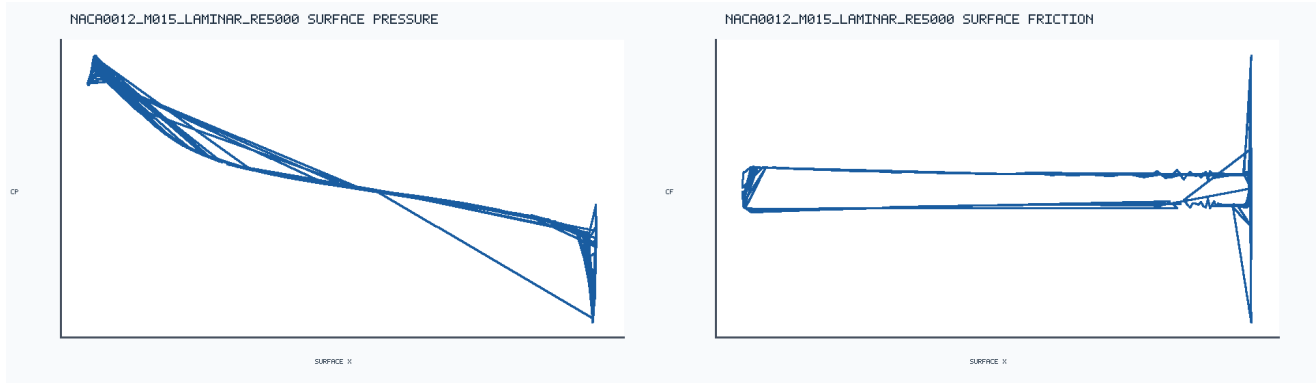


Figure 13: Surface pressure coefficient and skin-friction distributions for naca0012_m015_laminar_re5000.

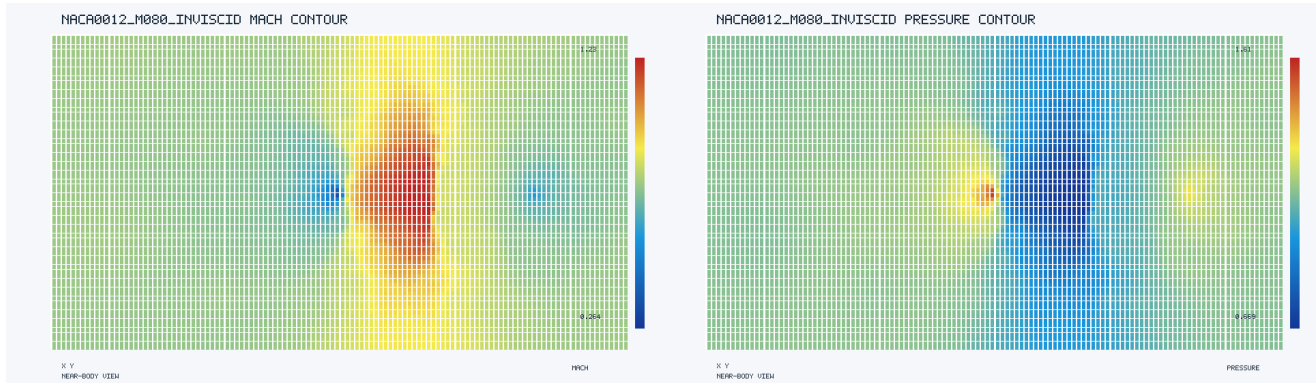


Figure 14: Final Mach and pressure contours for naca0012_m080_inviscid; variable names and view windows are shown in the renderings.

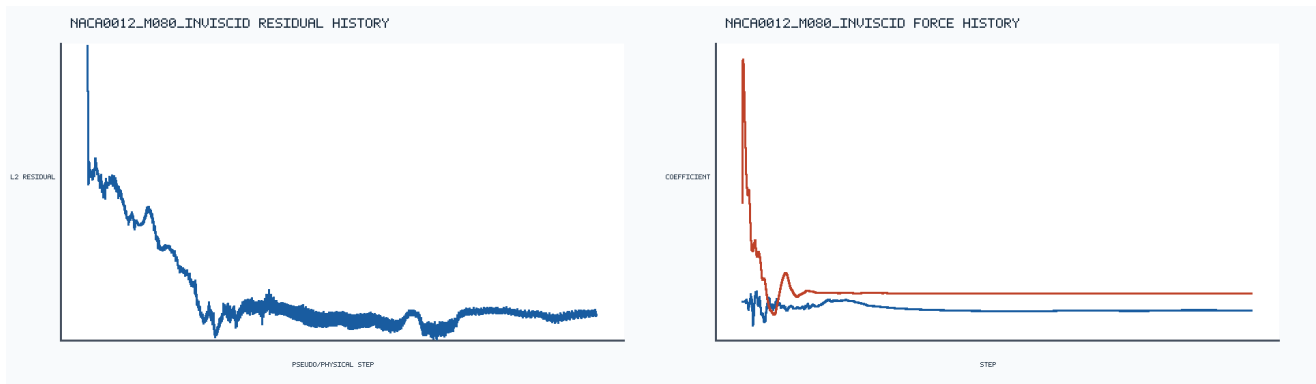


Figure 15: Residual and force histories for naca0012_m080_inviscid; source CSVs are listed in figure_manifest.csv.

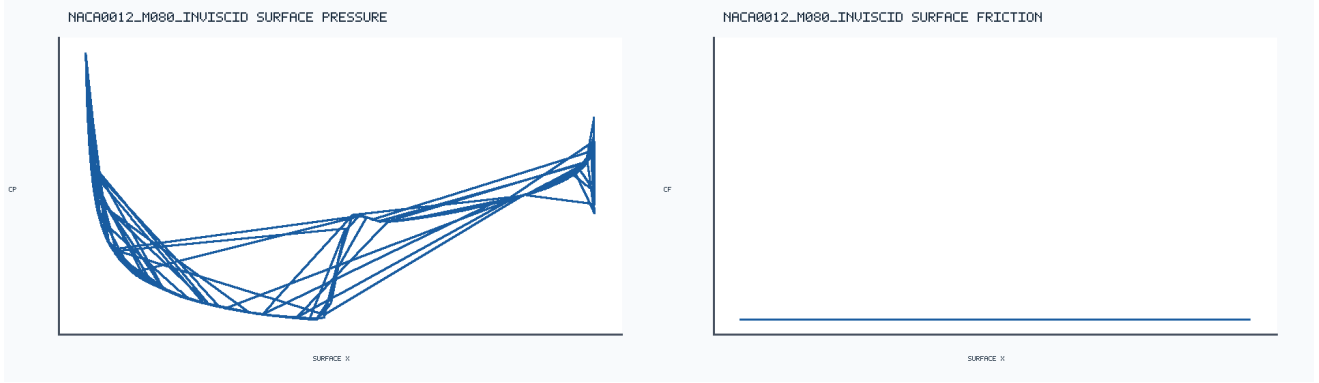


Figure 16: Surface pressure coefficient and skin-friction distributions for naca0012_m080_inviscid.

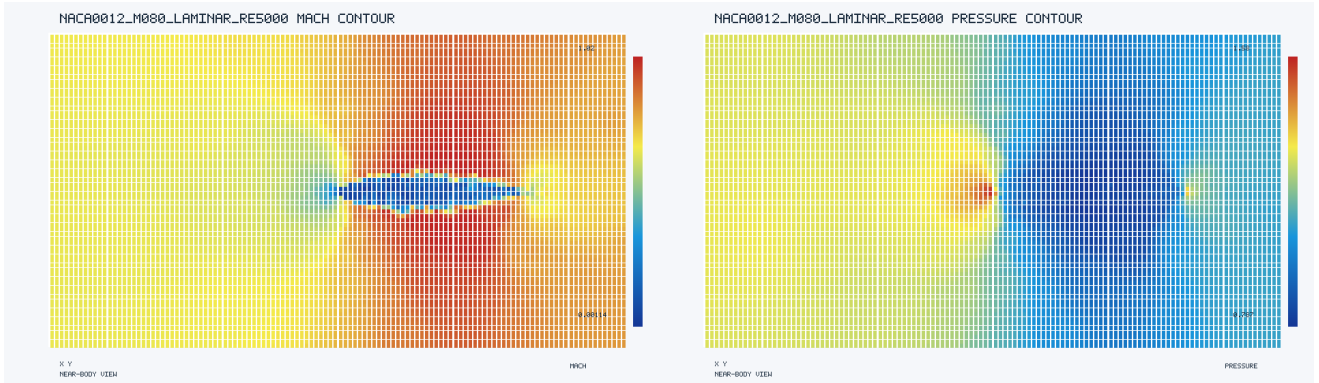


Figure 17: Final Mach and pressure contours for naca0012_m080_laminar_re5000; variable names and view windows are shown in the renderings.

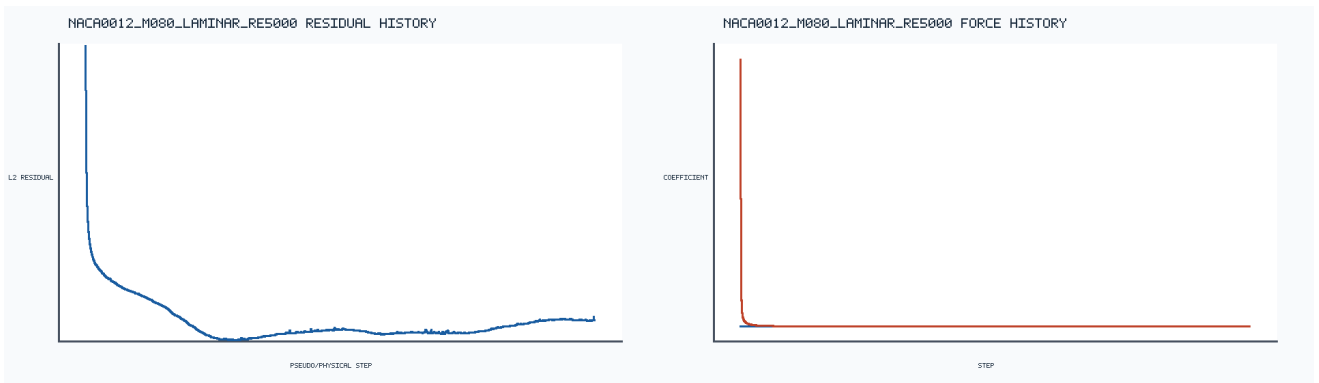


Figure 18: Residual and force histories for naca0012_m080_laminar_re5000; source CSVs are listed in figure_manifest.csv.

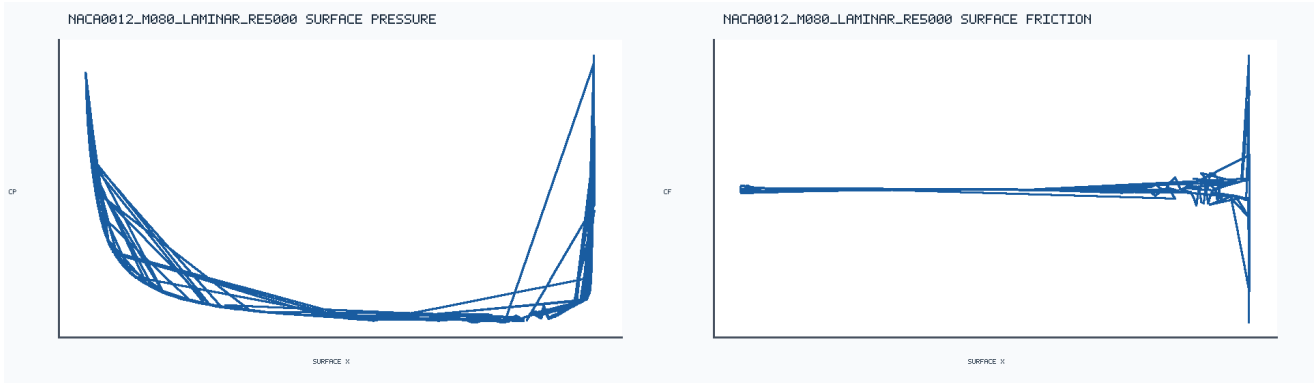


Figure 19: Surface pressure coefficient and skin-friction distributions for naca0012_m080_laminar_re5000.

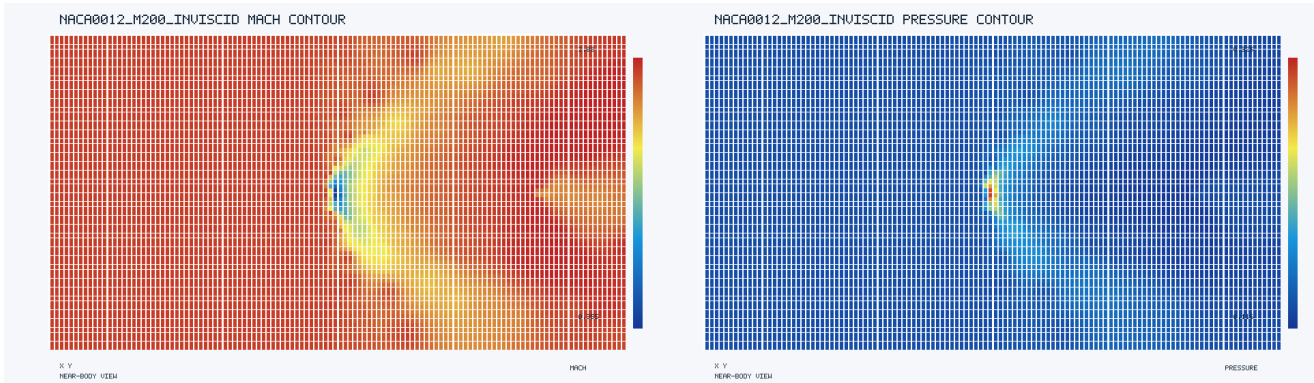


Figure 20: Final Mach and pressure contours for naca0012_m200_inviscid; variable names and view windows are shown in the renderings.

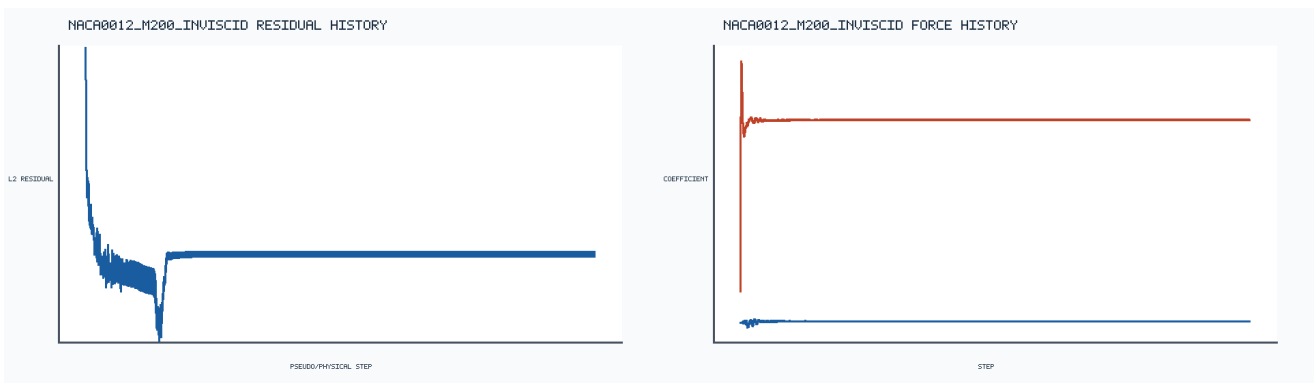


Figure 21: Residual and force histories for naca0012_m200_inviscid; source CSVs are listed in figure_manifest.csv.

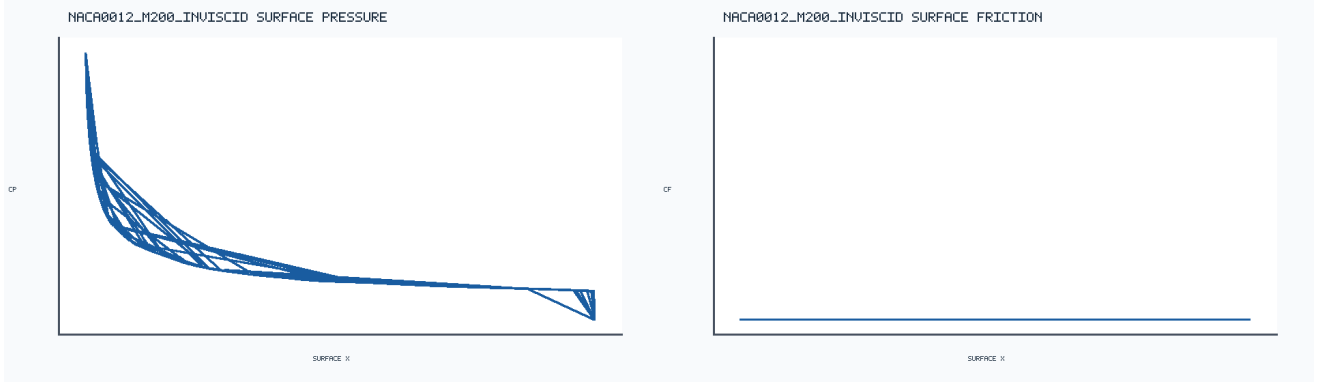


Figure 22: Surface pressure coefficient and skin-friction distributions for naca0012_m200_inviscid.

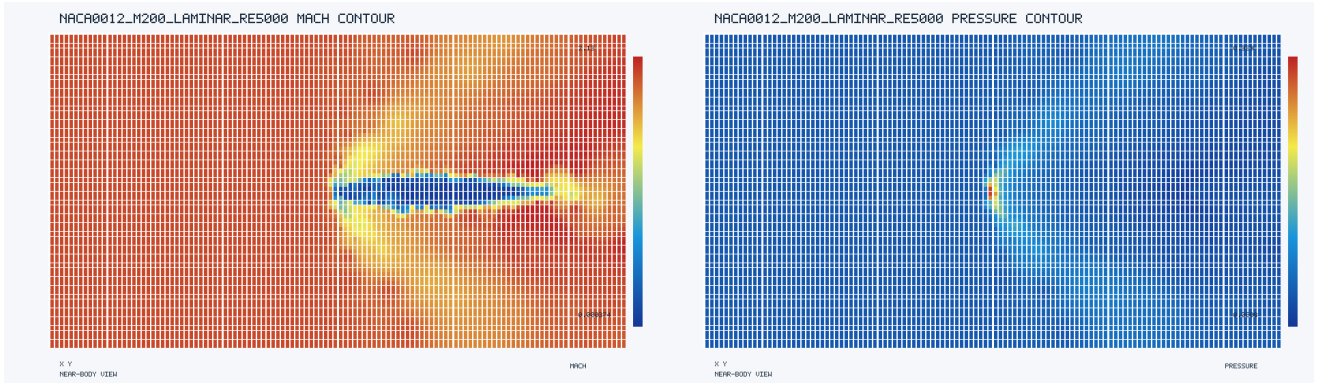


Figure 23: Final Mach and pressure contours for naca0012_m200_laminar_re5000; variable names and view windows are shown in the renderings.

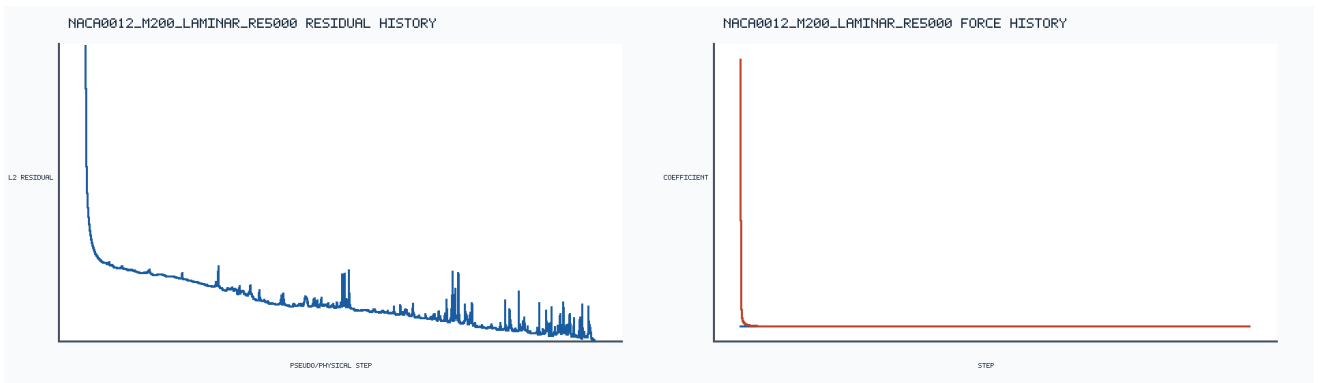


Figure 24: Residual and force histories for naca0012_m200_laminar_re5000; source CSVs are listed in figure_manifest.csv.

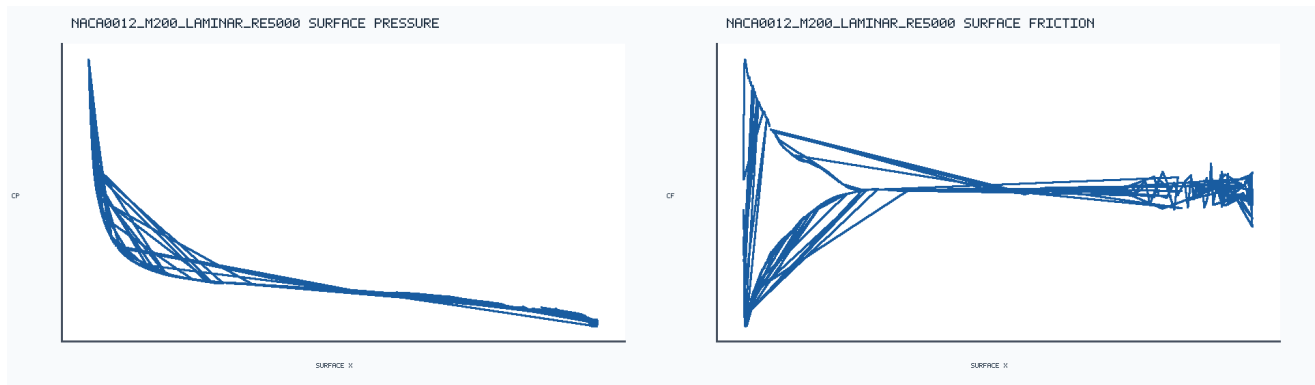


Figure 25: Surface pressure coefficient and skin-friction distributions for naca0012_m200_laminar_re5000.